

Basel Ahmed Sardah

Software Engineer and Computer Science graduate specializing in mobile application development using Flutter and Dart. Experienced in structured state management, local database optimization, and deploying applications to the Apple App Store. Capable of integrating pre-trained machine learning models into responsive client-side interfaces to deliver optimized, production-ready solutions.

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EDUCATION

King Faisal University

Bachelor of Computer Science - 4.79 / 5.00

2021 - 2026

WORK EXPERIENCE

Beaee Platform

Software Engineer Intern

Jan/2026 - Jul/2026

- **State Architecture:** Implemented Stacked architecture (MVVM) to strictly isolate UI code from state mutations, integrating GetIt to automate dependency injection for enhanced codebase maintainability.
- **Network Layer:** Engineered a robust REST network layer with automated token injection and 401 interceptors, reducing payload transmission latency by integrating client-side image compression prior to media uploads.
- **CI/CD & Deployment:** Configured cross-platform build pipelines using GitLab CI, eliminating manual compilation workflows and establishing automated release validation to TestFlight and the App Store.
- **DevOps & Parity:** Containerized core backend API services using Docker, achieving strict environment parity that eliminated integration bugs and friction between frontend and backend development teams.
- **Agile Collaboration:** Collaborated within an Agile team using Trello and Linear to deconstruct product requirements into granular tasks, stabilizing sprint delivery intervals across the SDLC.

PROJECT

Salad of Achievement: Gamified Productivity & Time-Tracking App

2025

- **Background Optimization:** Eliminated battery drain during background application termination by replacing continuous Isolate execution with an event-driven lifecycle model utilizing WidgetsBindingObserver.
- **State Architecture:** Managed state across 7 interactive screens using Provider and localized ValueNotifier layers, ensuring a continuous 60FPS fluidness during high-frequency UI mutations.
- **Database Engineering:** Architected a high-performance NoSQL data layer leveraging ObjectBox, utilizing Future.microtask() to eliminate asynchronous race conditions and setState() collisions during concurrent UI updates.

Predictive Analytics Web Dashboard: Edge Inference & Architecture

2025

- **Data Pipeline Optimization:** Developed an end-to-end machine learning pipeline in Python using Scikit-learn, Pandas, and NumPy, achieving a validated Mean Absolute Error (MAE) of less than 1%.
- **On-Device Inference:** Eliminated network overhead and server hosting costs by porting 19 pre-trained Linear Gradient Descent weights directly into a custom Dart inference engine, reducing inference latency to under 10ms.
- **Edge Architecture:** Built the responsive data analytics dashboard using Flutter Web and the Provider pattern to ensure decoupled state management across desktop and mobile browsers.
- **Data Visualization:** Integrated Syncfusion charts to map statistical score distributions, providing stakeholders with clear, localized prediction markers.

Wallet Expense Tracker: Local Ledger Application

2026

- **Memory Optimization:** Achieved O(1) read latency for real-time dashboard calculations by implementing caching strategy that atomically updates live financial aggregates within RAM during mutations.
- **Layered Architecture:** Architected a scalable MVVM structure in Flutter, using a centralized Data Access Object (DAO) pattern to cleanly isolate UI components from underlying SQLite database interactions.
- **Database Engineering:** Engineered a normalized SQLite relational schema utilizing temporal indexing, securing local data persistence against corruption through strict parameterized queries.
- **Performance Pipeline:** Sustained continuous 60FPS UI rendering by offloading heavy database I/O to asynchronous Dart Future queues, bypassing full-table scans through lightweight summary tables.

SKILLS

- Languages: Dart, Python, SQL (MySQL, SQLite, BigQuery)
- Frameworks & Mobile: Flutter (Android/iOS SDK, Flutter Web), FastAPI, Scikit-learn .
- Architecture & State Management: Provider, Stacked, MVVM, MVC, BLOC .
- Tools & DevOps: Git, GitHub, Docker, GitLab CI, TestFlight, Firebase, Agile/Scrum .